

S1 Table - extended tables

Model	Ensemble members
BCC-CSM2-MR	r1 r2 r3
BCC-ESM1	r1 r2 r3
CAMS-CSM1-0	r1 r2
CanESM5	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
CNRM-CM6-1	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
CNRM-CM6-1-HR	r1
CNRM-ESM2-1	r1 r2 r3 r4 r5
E3SM-1-1	r1
EC-Earth3	r1 r2 r4 r7
EC-Earth3-Veg	r1 r2 r3 r4
FIO-ESM-2-0	r1
INM-CM4-8	r1
INM-CM5-0	r1 r2 r5 r6 r7 r8 r9 r10
IPSL-CM6A-LR	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
MIROC6	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
HadGEM3-GC31-LL	r1 r2 r3 r4
HadGEM3-GC31-MM	r1 r2
UKESM1-0-LL	r2 r3 r4 r5 r6 r7 r8 r9 r10
MPI-ESM1-2-HR	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
MRI-ESM2-0	r1 r5
GISS-E2-1-G	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
CESM2	r1 r2 r3 r4 r5 r6 r7 r8 r9 r10
CESM2-WACCM	r1 r2 r3
NorESM2-LM	r1 r2 r3
GFDL-CM4	r1
NESM3	r1 r2 r3 r4 r5
SAM0-UNICON	r1

Table A: List of models and their ensemble members that are used in this study. All ensemble members for a listed rx are rxilp1f1 except for CNRM-CM6-1, CNRM-ESM2-1, and UKESM1-0-LL which are rxilp1f2 and HadGEM3-GC31-MM HadGEM3-GC31-LL which are rxilp1f3.

	S26.5	S35	SSTI	Any
Any	24.0	35.0	21.0	80.0
One	15.0	21.0	18.0	54.0
Two	7.0	11.0	3.0	21.0
Three	2.0	3.0	0.0	5.0

Table B: Same as Table A in S2 Appendix for a window size of 50 years.

	S26.5	S35	SSTI	Any
Any	31.0	40.0	14.0	85.0
One	17.0	23.0	7.0	47.0
Two	11.0	10.0	6.0	27.0
Three	3.0	7.0	1.0	11.0

Table C: Same as Table A in S2 Appendix for a window size of 70 years.

	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10
BCC-CSM2-MR										
BCC-ESM1										
CAMS-CSM1-0		S26.5								
CanESM5			S35		S35		S26.5 S35	S26.5	S35	S26.5 ASSTI
CNRM-CM6-1								S26.5		
CNRM-CM6-1-HR										
CNRM-ESM2-1										
E3SM-1-1										
EC-Earth3		S35								
EC-Earth3-Veg										
FIO-ESM-2-0	S35									
INM-CM4-8										
INM-CM5-0		S26.5								
IPSL-CM6A-LR										
MIROC6					S35	S35	S26.5	S26.5 S35		
HadGEM3-GC31-LL			S26.5	S35						
HadGEM3-GC31-MM	S35									
UKESM1-0-LL					S35			S35		
MPI-ESM1-2-HR										ASSTI
MRI-ESM2-0										
GISS-E2-1-G		ASSTI	ASSTI							
CESM2			S26.5 S35	S35			S26.5	S26.5 S35	S35	
CESM2-WACCM			S35							
NorESM2-LM			S26.5							
GFDL-CM4										
NESM3				S26.5						
SAM0-UNICON										

Table D: Significant increases of λ . The same abbreviations for the time series as Fig 6 in the main text are used. For each ensemble member the time series in which there is a significant increase is listed in the corresponding cell. Note that not all models have all 10 ensemble members - the list of members used can be seen in Table A in S1 Table.

	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10
BCC-CSM2-MR			S35							
BCC-ESM1										
CAMS-CSM1-0										
CanESM5				ASSTI			S35	S26.5	S35	S26.5 ASSTI
CNRM-CM6-1								S26.5 ASSTI		
CNRM-CM6-1-HR										
CNRM-ESM2-1										
E3SM-1-1										
EC-Earth3										
EC-Earth3-Veg										
FIO-ESM-2-0	S35									
INM-CM4-8										
INM-CM5-0	S26.5						S26.5			
IPSL-CM6A-LR										
MIROC6	S35 ASSTI	S26.5 S35			S35			S35		
HadGEM3-GC31-LL		S35 ASSTI	S26.5 S35	S26.5 S35 ASSTI						
HadGEM3-GC31-MM	S35									
UKESM1-0-LL		S26.5						S35		S26.5
MPI-ESM1-2-HR										
MRI-ESM2-0										
GISS-E2-1-G		S26.5		ASSTI						
CESM2	S26.5		S26.5 S35			S26.5	S35	S26.5 S35		
CESM2-WACCM										
NorESM2-LM			S26.5							
GFDL-CM4										
NESM3				S26.5						
SAM0-UNICON										

Table E: Significant increases in the autocorrelation. Same structure as Table D in S1 Table.

	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10
BCC-CSM2-MR										
BCC-ESM1			S26.5 S35 ASSTI							
CAMS-CSM1-0										
CanESM5										
CNRM-CM6-1								S26.5 S35 ASSTI		
CNRM-CM6-1-HR										
CNRM-ESM2-1		S26.5								
E3SM-1-1										
EC-Earth3										
EC-Earth3-Veg		S26.5 S35								
FIO-ESM-2-0	S35									
INM-CM4-8										
INM-CM5-0		S26.5					S26.5		S35	
IPSL-CM6A-LR				S35			ASSTI	S26.5 S35		
MIROC6	ASSTI		S35							
HadGEM3-GC31-LL				S35						
HadGEM3-GC31-MM										
UKESM1-0-LL								S35		
MPI-ESM1-2-HR										
MRI-ESM2-0										
GISS-E2-1-G		ASSTI	S35		ASSTI					
CESM2		S26.5						S35	S35	
CESM2-WACCM	S26.5 S35									
NorESM2-LM										
GFDL-CM4										
NESM3										
SAM0-UNICON										

Table F: Significant increases in the variance. Same structure as Table D in S1 Table.